

SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 2.1: Properties of Waves — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.1: Properties of Waves
Driving Question	DRIVING QUESTION: How do sound waves reflect, and why do some places produce stronger echoes than others?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon: Echoes at Hell's Gate, Ngong, Kisumu and Nairobi	At Hell's Gate (85 m) and Ngong Hills (170 m) we heard clear echoes after a delay (0.50 s and 1.00 s). In the Kisumu hall (12 m) the sound was loud, muddy and drawn-out. On the open Nairobi grounds (nearest hard wall 300 m) we heard no echo. At the school wall we also listened for our own sound coming back and described what we heard.	An echo is sound coming back to us after it meets a surface. Different places give different results, and there is a delay between the shout and the echo. We do not yet know exactly why some places give strong echoes and others do not.	Our first ideas: the sound travels from us to the wall or cliff and comes back, and the delay may be the time it takes to travel there and back. The hall may sound muddy because of many walls, and the open field may give no echo because the sound spreads out or is absorbed. We will test these ideas in later lessons.
Lesson 2 Rectilinear Propagation: Light and Wave Paths from Source to Cliff and Back	We saw that shadows have sharp edges shaped like the object, that light passes through aligned holes but not when one hole is moved, that a pinhole camera makes an upside-down image, and in the PhET Geometric Optics simulation that the ray from the source stays straight until it meets an obstacle. We saw that light seems instant while a shout at Hell's Gate takes 0.50 s to return.	Light travels in straight lines in a uniform medium (rectilinear propagation). A ray is a straight line with an arrow showing the direction in which wave energy travels. Light travels at 3.0×10^8 m/s, about a million times faster than sound at 340 m/s, so sound shows a delay over distance.	The shout at Hell's Gate can be drawn as a straight ray from the learner to the cliff 85 m away and a straight ray back, a total of 170 m. At 340 m/s this takes 0.50 s, which is the delay Wanjiru and Kiprono noticed. We still need to find out what happens at the cliff, why the sound returns, and why the echo is strong at some places and weak at others.
Lesson 3 Reflection of Waves: The Laws of Reflection	We shone a ray at a plane mirror and found that when $i = 40$ degrees, $r = 40$ degrees (measured from the normal). We watched ripples bounce off a barrier at an equal angle, and drew a sound ray reflecting from the Hell's Gate cliff.	The two laws of reflection: the angle of incidence equals the angle of reflection (both measured from the normal), and the incident ray, reflected ray and normal lie in the same plane. These laws apply to light, water waves and sound.	When Wanjiru shouts at the Hell's Gate cliff, the sound travels in a straight line, reflects from the cliff obeying the law of reflection ($i = r$) and returns to the listeners. This explains how the voice comes back, but not yet the delay or why some places give

			stronger echoes than others.
Lesson 4 Reflection of Sound: Echo Measurement and Surfaces	We observed that sound clapped at about 50 m from the school wall returned after about 0.294 s per interval (10 intervals in 2.94 s), that the delays at the four sites were 0.50 s (Hell's Gate), 1.00 s (Ngong Hills), 0.07 s (Kisumu hall) and 1.76 s (Nairobi grounds), and that hard surfaces such as rock (90%) and painted concrete (85%) return much more sound than curtains (15%) or grass and foam (10%).	We learned that $d = vt/2$ gives the one-way distance from the round trip time, that a distinct echo needs a delay of at least 0.1 s which means a reflector at least 17 m away, and that strong echoes need hard smooth surfaces while soft surfaces absorb sound. Reverberation is overlapping reflections that merge and prolong the sound.	We can explain that the voice returns after a delay of $2d/v$ because sound reflects from the cliff, valley wall or building. Echoes are clear at Hell's Gate and Ngong Hills because the surfaces are far (more than 17 m) and hard; the Kisumu hall is muddy because the wall is only 12 m away (0.07 s) and many hard surfaces keep reflecting; and the Nairobi field gives no echo because the sound spreads out, grass absorbs it and the only big wall is 300 m away, so the return is too faint.
Lesson 5 Refraction of Waves: Why Waves Change Direction When Their Speed Changes	We observed that a straw standing in water looks bent or broken at the water surface, a coin in a cup appears when water is poured in, and a Lake Victoria pool of real depth 1.33 m looks only 1.00 m deep. In the ripple tank videos, water waves moving from deep to shallow water changed direction and had shorter wavelengths. We saw no bend when we looked straight down along the normal.	We learned that refraction is the change in direction of a wave when it passes from one medium to another because its speed changes. Light travels at 3.0×10^8 m/s in air, 2.26×10^8 m/s in water and 2.0×10^8 m/s in glass, so it slows down and bends toward the normal when entering water or glass. Sound travels at 340 m/s in air and about 1500 m/s in water, so it speeds up. Reflection keeps the wave in the same medium ($i = r$), while refraction sends it into the second medium.	The straw looks broken and the pool looks shallower (1.00 m instead of 1.33 m) because light changes speed and direction at the water surface, so rays from the bottom bend away from the normal and our eyes trace them back to a higher apparent position. This adds to the echo model: at a boundary, part of the wave is reflected (the echo we hear at Hell's Gate after 0.50 s and Ngong Hills after 1.00 s) and the rest is absorbed or goes into the second medium and refracts. The more energy that is reflected (cliff rock 90%) the stronger the echo, and the more that is absorbed (grass 10%) the weaker the echo.
Lesson 6 Snell's Law, Refractive Index and Lenses: Bending Waves in Our Echo Model	Light bent towards the normal when it entered water (r about 32 degrees) and glass (r about 28 degrees) from air at $i = 45$ degrees; a converging lens brought parallel rays together at a focal point and a diverging lens spread them out.	Snell's law $n_1 \sin i = n_2 \sin r$ with $n = 1.00$ (air), 1.33 (water) and 1.50 (glass); a higher n means slower light (2.26×10^8 m/s in water, 2.0×10^8 m/s in glass) and more bending towards the normal; converging and diverging lenses are used in spectacles, cameras and microscopes.	At a boundary a wave can reflect and also refract. The Hell's Gate echo (0.50 s) and the Ngong Hills echo (1.00 s) come from reflection, which sends the sound back. Refraction describes the part of the wave that enters a new medium and bends because its speed changes. Refraction alone does not explain the echo, so the muddy Kisumu hall sound and the missing echo on the Nairobi grounds still need other wave ideas.
Lesson 7 Diffraction of Waves: Why Sound Turns Corners but Light Does Not	We observed that waves passing through a gap spread out on the far side, and that the spreading was greatest when the gap was about the same size as the wavelength. In the simulation, narrow gaps and long	We learned that diffraction is the spreading of waves around edges and through gaps. We learned to calculate wavelength using $v = f \times \text{wavelength}$: 340 Hz gives 1.0 m and 1000 Hz gives 0.34 m, while green light is	Sound turns corners more than light because its wavelength (about 1.0 m for 340 Hz) is comparable to the size of doorways and obstacles, so it spreads strongly, while the wavelength of light (5.5 x

	wavelengths gave strong spreading, while wide gaps gave a nearly straight beam. At the classroom door, we could hear Mr. Mwangi around the corner but could not see him, and the torch light made a sharp patch.	about 5.5×10^{-7} m. A 1.0 m doorway is comparable to the 340 Hz sound wavelength but enormously larger than the wavelength of light. Diffraction does not change frequency or wavelength.	10^{-7} m) is tiny compared to the gap, so light barely bends and seems to travel in straight lines. This adds to our echo model: the shout spreads as it travels as well as reflecting, which we will test later against the Kisumu hall and the open Nairobi grounds.
Lesson 8 Interference of Waves: When Sound Waves Meet in the Kisumu Hall	In the ripple tank and in the PhET Sound simulation with two speakers at 340 Hz, we observed alternating regions of large movement or loud sound and regions of almost no movement or quiet sound. At path difference 1.0 m ($\lambda = 1.0$ m) the sound was loud and at 0.5 m it was quiet.	Waves that meet add their displacements (superposition). Whole-wavelength path differences give constructive interference (loud) and half-wavelength path differences give destructive interference (quiet). Noise-cancelling headphones use a wave half a wavelength out of step with the noise.	In the Kisumu hall (site C) the far wall is 12 m away, the delay is 0.07 s (less than 0.1 s), so reflections overlap the voice and each other. They add in some places and cancel in others, and the many hard surfaces keep the sound lingering for about 1.2 s, so it is loud, muddy and drawn out. At Hell's Gate there is a single clean reflection after 0.50 s, so the echo is distinct.
Lesson 9 Stationary Waves and Resonance: Nodes, Antinodes and the Pitch of the Nyatiti and Flute	We observed that a plucked or shaken string fixed at both ends (and an air column in a tube) forms steady patterns at particular frequencies only. In these patterns some points never move (nodes) and some swing the most (antinodes). The fundamental has 2 nodes (the ends) and 1 antinode; the second harmonic has 3 nodes and 2 antinodes. A shorter string or tube gave a higher pitch, and the air column became loud at a particular length (resonance).	We learned that a stationary wave forms when a wave and its reflection travel in opposite directions and interfere: they add at antinodes and cancel at nodes. For the fundamental, wavelength = $2L$, so $f = v/2L$. Pitch depends on frequency, and resonance happens when a driving frequency matches a natural frequency of the string or air column.	We can explain that the nyatiti string 0.60 m long with wave speed 120 m/s has wavelength 1.20 m and fundamental $f = 120/1.20 = 100$ Hz, with a second harmonic of 200 Hz, and that a flute modelled as a 0.34 m open tube has wavelength 0.68 m and $f = 340/0.68 = 500$ Hz. Reflection at the ends plus interference gives stationary waves, and in an echo situation such as the Kisumu hall reflected sound can also build up at some frequencies, which adds to our model of why some places sound stronger than others.
Lesson 10 Doppler Effect: Why the Ambulance Siren Changes Pitch as It Passes	We watched and acted out a moving sound source. The pitch of the ambulance siren sounded higher as it approached (744 Hz) and lower as it receded (661 Hz), although the siren emits 700 Hz.	Relative motion of source and observer changes the observed frequency: waves bunch up ahead of a moving source (higher frequency) and stretch out behind it (lower frequency). The speed of sound stays 340 m/s and $v = f \times \text{wavelength}$ still holds. This is used in radar to measure speed and in astronomy to tell whether stars and galaxies are moving towards or away from us.	At Hell's Gate and Ngong Hills the sound returns from a fixed cliff, so its pitch matches the shout and only the delay (0.50 s and 1.00 s) tells us the distance. If the source, observer or reflecting surface were moving, the returned sound would also change in frequency, which is how radar guns find the speed of a matatu. Doppler explains pitch change, not why the Kisumu hall is muddy or why the Nairobi grounds have no echo.
Lesson 11 Waves in Communication and Medicine: FM Radio and	We observed that the 91.0 MHz FM carrier is a radio wave of wavelength 3.30 m travelling at 3.0×10^8 m/s, and that in FM the voice changes the frequency of the	We learned that radio waves are electromagnetic waves that need no medium, that FM carries sound by changing the carrier frequency, and that ultrasound	We explain that the same wave ideas explain the phenomenon and the technologies: a wave travels, meets a boundary, reflects, and returns after a time

Ultrasound as Echo Applications	carrier while its amplitude stays constant. We observed that a 3.5 MHz ultrasound pulse in soft tissue (1540 m/s) has a wavelength of 0.44 mm and that an organ boundary 7.7 cm deep returns an echo after 100 microseconds.	imaging uses reflected pulses and echo timing with $d = vt/2$ to locate tissue boundaries. We learned that the echo idea from Hell's Gate (0.50 s) and Ngong Hills (1.00 s) works at both large and tiny scales.	equal to $2d/v$, so the delay tells us the distance. In the radio, the frequency of the wave carries the information. This shows how wave properties are applied in day-to-day life, and we will use all these ideas in Lesson 12 to give the final explanation of the four echo sites.
Lesson 12 Noise Pollution and the Final Model: Why Do Some Places Echo and Others Do Not?	We observed that noise levels range from 30 dB (whisper) to 95 dB (matatu with loud music), with 85 dB as the class safe limit for up to 8 hours. We also observed that hard materials such as painted concrete (85% reflected) and wooden board (60%) reflect sound strongly, while curtains (15%) and foam or grass (10%) reflect very little. At the four sites, we saw that Hell's Gate (0.50 s) and Ngong Hills (1.00 s) give clear echoes, the Kisumu hall (0.07 s) gives a loud muddy sound, and the Nairobi open grounds give no echo.	We learned that noise above safe limits can damage hearing and cause stress and poor concentration, and that absorbing materials and barriers reduce noise and reverberation. We learned that an echo needs a reflector at least 17 m away (delay of at least 0.1 s) and a surface that reflects enough energy, that $d = vt/2$ with $v = 340$ m/s, and that reverberation is overlapping reflections that merge and prolong the sound. We also learned how reflection, refraction, diffraction, interference, resonance and the Doppler effect all follow the same wave ideas.	Sound is a mechanical wave that leaves the shouting learner, reflects from a large hard smooth surface following the law of reflection ($i = r$ from the normal) and returns after $t = 2d/v$. At Hell's Gate and Ngong Hills the surface is far away (more than 17 m) and hard, so a strong separate echo is heard. In the Kisumu hall the wall is only 12 m away (0.07 s) with many hard surfaces, so reflections overlap the voice, causing loud, muddy reverberation of about 1.2 s, which curtains and foam can reduce by absorbing energy. On the open Nairobi grounds the sound spreads out and is absorbed by grass, and the only big wall is 300 m away (1.76 s), so the returning sound is too faint to hear.

ATTRIBUTION AND LICENSING

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